

Error in shared pool

```
Errors in file /u01/app/oracle/diag/rdbms/mustbild/MUSTBILD/trace/MUSTBILD_ora_2135556.trc (incident=1283553):
ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")
Use ADRCI or Support Workbench to package the incident.
See Note 411.1 at My Oracle Support for error and packaging details.
2026-03-17T12:47:49.041845+02:00
Errors in file /u01/app/oracle/diag/rdbms/mustbild/MUSTBILD/trace/MUSTBILD_ora_2135558.trc (incident=1283561):
ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")
Use ADRCI or Support Workbench to package the incident.
See Note 411.1 at My Oracle Support for error and packaging details.
2026-03-17T12:47:51.721387+02:00
Errors in file /u01/app/oracle/diag/rdbms/mustbild/MUSTBILD/trace/MUSTBILD_ora_2135560.trc (incident=1283569):
ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")
Use ADRCI or Support Workbench to package the incident.
See Note 411.1 at My Oracle Support for error and packaging details.
2026-03-17T12:47:54.406170+02:00
Errors in file /u01/app/oracle/diag/rdbms/mustbild/MUSTBILD/trace/MUSTBILD_ora_2135562.trc (incident=1283577):
ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")
Use ADRCI or Support Workbench to package the incident.
See Note 411.1 at My Oracle Support for error and packaging details.
2026-03-17T12:47:57.091663+02:00
Errors in file /u01/app/oracle/diag/rdbms/mustbild/MUSTBILD/trace/MUSTBILD_ora_2135566.trc (incident=1283593):
ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")
Use ADRCI or Support Workbench to package the incident.
See Note 411.1 at My Oracle Support for error and packaging details.
2026-03-17T12:47:59.779545+02:00
Errors in file /u01/app/oracle/diag/rdbms/mustbild/MUSTBILD/trace/MUSTBILD_ora_2135576.trc (incident=1283601):
ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")
Use ADRCI or Support Workbench to package the incident.
See Note 411.1 at My Oracle Support for error and packaging details.
2026-03-17T12:48:02.457704+02:00
Errors in file /u01/app/oracle/diag/rdbms/mustbild/MUSTBILD/trace/MUSTBILD_ora_2135579.trc (incident=1283609):
ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")
Use ADRCI or Support Workbench to package the incident.
See Note 411.1 at My Oracle Support for error and packaging details.
```

ORA-04031: unable to allocate 16272 bytes of shared memory ("shared pool","unknown object","sga heap(2,0)","ksu:stats_freelist")

ORA-04031: Oracle tried to allocate memory in the Shared Pool but failed.

16272 bytes: the amount of memory it tried to allocate.

"shared pool": the memory area where the allocation failed.

"unknown object": the exact object type Oracle was trying to allocate is unknown.

"sga heap(2,0)": the specific heap within the SGA where the problem occurred.

"ksu:stats_freelist": the internal structure or object type related to statistics or temporary objects that needed memory.

In short: Oracle needed ~16 KB more memory in the Shared Pool but there wasn't enough free space → this caused ORA-04031.

Automatic Shared Memory Management (ASMM):

- ASMM is active, so Oracle dynamically manages the size of the Shared Pool within the SGA limits.
- Despite ASMM, this error can occur if a request for memory exceeds the available free memory before Oracle can expand the pool.

--check memory

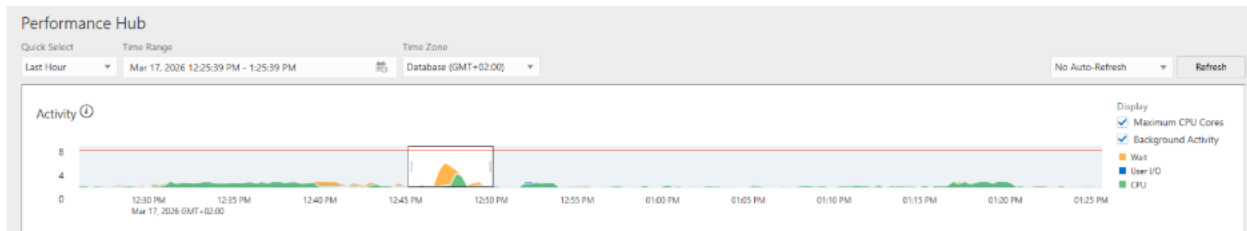
	total	used	free	shared	buff/cache	available
Mem:	47Gi	20Gi	1.7Gi	22Mi	25Gi	26Gi
Swap:	15Gi	925Mi	15Gi			

--the safest Memory size is 48GB

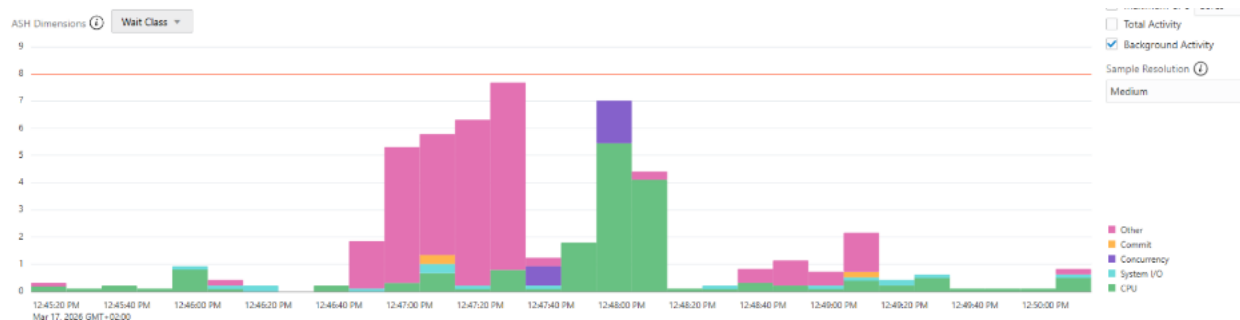
From OEM

select DB -> performance hup -> ASH Analytics

-- Activity



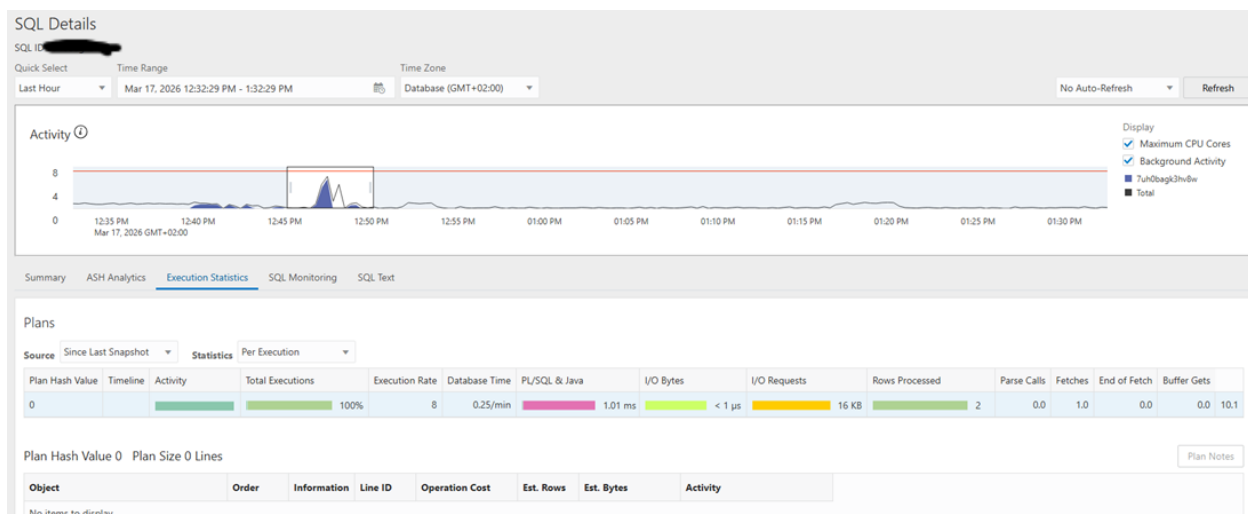
-- Average Active Sessions



Wait Class -> SQL Details

SQL ID ▾	by Wait Class	Tune SQL	Create SQL Tuning Set	Columns ▾	
SQL ID	Activity (Average Active Sessions)	SQL Plan Hash Value	SQL Type	Service	
[REDACTED]	 0.89	N/A	PL/SQL EXECUTE	[REDACTED]	
[REDACTED]	 0.08	27270497	SELECT	[REDACTED]	SQL ID: [REDACTED], SQL Text: BEGIN dbms_session.set_role(:1); END;
[REDACTED]	 0.03	N/A	PL/SQL EXECUTE	[REDACTED]	
[REDACTED]	 0.03	3752461848	SELECT	[REDACTED]	
[REDACTED]	 0.02	N/A	PL/SQL EXECUTE	[REDACTED]	
[REDACTED]	 < 0.01	1911072918	SELECT	SYSSBACKGROUND	

BEGIN dbms_session.set_role(:1); END;




```
SQL> SELECT name,
        ROUND(bytes/1024/1024,2) AS size_mb
FROM v$sgastat
WHERE pool='shared pool' AND name='free memory'; 2    3    4

NAME                                SIZE_MB
-----
free memory                        181

SQL> SELECT name,
        ROUND(bytes/1024/1024/1024,4) AS size_gb
FROM v$sgastat
WHERE pool='shared pool'
ORDER BY bytes DESC
FETCH FIRST 10 ROWS ONLY; 2    3    4    5    6

NAME                                SIZE_GB
-----
ksunfy_meta 1                      .3732
private strands                    .2098
free memory                        .1720
session                            .1429
db_block_hash_buckets              .1250
SQLA                                .1209
KTI-UNDO                           .1202
S0 private sga                     .1196
KGLH0                               .0625
KSRMA State Object                 .0594

10 rows selected.

SQL> █
```

svr-dbb-03.must.edu.eg 2% 20.74 GB / 47.21 GB 5.17 Mb/s

--Shared Pool current size: ~2.375 GB

--Free memory: ~262 MB

COLUMN pool FORMAT A15

COLUMN name FORMAT A30

COLUMN current_size_gb FORMAT 999.9999

COLUMN min_size_gb FORMAT 999.9999

COLUMN max_size_gb FORMAT 999.9999

COLUMN size_gb FORMAT 999.9999

-- **1** Shared Pool sizes

```
SELECT component AS pool, ROUND(current_size/1024/1024/1024,4) AS current_size_gb, ROUND(min_size/1024/1024/1024,4) AS min_size_gb, ROUND(max_size/1024/1024/1024,4) AS max_size_gb FROM v$sga_dynamic_components WHERE component='shared pool';
```

```
-- 2 أكبر 10 عناصر مستهلكة للذاكرة داخل Shared Pool SELECT name, ROUND(bytes/1024/1024/1024,4) AS size_gb FROM v$sgastat WHERE pool='shared pool' ORDER BY bytes DESC FETCH FIRST 10 ROWS ONLY;
```

```
-- 3 free memory داخل Shared Pool SELECT name, ROUND(bytes/1024/1024,2) AS size_mb FROM v$sgastat WHERE pool='shared pool' AND name='free memory';
```